Research Proposal: In-Depth Analysis of Homomorphic Encryption Libraries

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Introduction

Homomorphic Encryption (HE) is a cryptographic technique that allows computations to be performed directly on encrypted data. In a regular encryption scheme, data must usually be decrypted before it can be processed. This creates a security gap: even if the data is protected while stored or transmitted, it may still be exposed during computation. Homomorphic encryption addresses this problem by enabling a server or cloud service to compute over ciphertexts without learning the underlying plaintext values.

The basic idea is that encrypted values can be manipulated in a way that corresponds to meaningful operations on the original data. For example, if two encrypted numbers are added together, the decrypted result should match the addition of the original plaintext numbers. Some HE schemes support only limited operations, such as addition or multiplication, while more advanced schemes support deeper computations by combining multiple operations. This makes HE relevant for privacy-preserving analytics, secure outsourcing of computation, healthcare data processing, financial analysis, and other cases where sensitive data should remain hidden from the computing environment.

In this project, we focus on the practical use of homomorphic encryption libraries. In particular, we propose to compare two HE implementations, **TenSEAL** and **OpenFHE**, through a small set of basic encrypted computations. The focus of the project is to understand the practical cost of using HE libraries, especially in terms of runtime, memory usage, implementation complexity, and limitations.

This document is a research proposal for the Confidential Computing course at Ben-Gurion University of the Negev.

Research Goal

The main goal of this project is to compare the computational overhead of two homomorphic encryption libraries, **TenSEAL** and **OpenFHE**, in terms of runtime and memory usage.

The research question is:

> *What is the practical overhead of performing basic computations using two different homomorphic encryption implementations, compared to equivalent plaintext computation?*

The comparison will focus on simple operations that are common in data analysis, such as addition, multiplication, summation, and average calculation. The goal is to provide a practical view of how expensive these operations become when performed over encrypted data.

Research Plan

The project will begin with a short technical study of homomorphic encryption and the selected libraries. We will examine the basic concepts required to run encrypted computations, including key generation, encryption, decryption, ciphertext operations, and the difference between exact and approximate arithmetic where relevant.

After this initial study, we will set up both libraries and implement a common set of basic computations. The mandatory implementation part will include:

- Key generation.
- Encryption and decryption of numeric values.
- Encrypted addition.
- Encrypted multiplication.
- Encrypted summation over a small vector of values.
- Encrypted average calculation.
- Equivalent plaintext computations for baseline comparison.

For each library, we plan to measure:

- Runtime of encryption, computation, and decryption.
- Memory usage during the computation.
- Size of encrypted values, where practical.
- Ease of implementation and usability.

- Practical limitations encountered during development.

The main comparison will be between plaintext computation, TenSEAL-based encrypted computation, and OpenFHE-based encrypted computation. The results will be presented using tables and, if useful, simple graphs.

The following extensions will be considered optional, depending on time and on the findings during implementation:

- Encrypted dot product.
- Encrypted weighted score calculation.
- Additional extensions, such as testing larger input sizes, different parameter settings, or additional operations.

The plan may change according to the findings during the project. In particular, the exact benchmark structure and the optional extensions may be adjusted according to the practical behavior of the libraries and the difficulty of implementing comparable operations in both.

As a possible real-world use case, we will discuss privacy-preserving healthcare analytics. For example, homomorphic encryption could allow a hospital, research institution, or cloud provider to compute simple statistics over sensitive patient data without exposing individual patient records. This use case will serve as a motivating example for the discussion, but the core project will focus on comparing the two HE implementations through basic benchmark operations.

Expected Deliverables

The expected deliverables of the project are:

- A working implementation demonstrating basic homomorphic encryption operations using TenSEAL and OpenFHE.
- A small benchmark comparing encrypted computation with equivalent plaintext computation.
- Runtime and memory measurements for the selected operations.
- A short discussion of a possible healthcare analytics use case and how HE could support privacy-preserving computation in such a setting.
- A final report describing the implementation, experiments, results, and practical conclusions.
- A final presentation summarizing the project goal, methodology, findings, and limitations.

The final outcome should provide a practical understanding of the cost and usability of homomorphic encryption libraries. In particular, the project aims to show not only that encrypted computation is possible, but also what overheads and limitations appear when trying to use it for basic computation tasks.